

The HyMeKo topology zoo

generated architectures as walk bases — a reference

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2026-06-28 (JST)

Abstract

A registry of signed-graph topologies over matched vertices, for the isomorphic-controllers program (“which topology controls which plant”). Each topology is a candidate **walk basis**: SA-HSiKAN gathers along the signed L -hop walk-holonomy B^L , so a topology *generates* the controller’s structural prior. Three groups: **generic** graphs, combinatorial **designs**, and the **extremal** high-symmetry constructions. Each builder is an edge-list through `topology_zoo.signed_graph`; the designs come from `hypergraph_designs`. Invariants per topology (the walk-structure measures) are in `topology_invariants`.

1 The families

group	topology	construction	note
generic	chain	line $0-1-\dots$	low connectivity
	ring	cycle	regular, balanced
	star	one hub $\rightarrow$ all	degenerate (max overlap)
	tree	balanced binary tree	hierarchical
	grid	$\sqrt{N} \times \sqrt{N}$ 4-neighbour	needs square N
	small_world	ring + random shortcuts	Watts–Strogatz
	random	$G(n, p)$ + spanning chain	connected random
	complete	all pairs	densest
design	steiner9	Steiner triple system $S(2, 3, 9)$	balanced 2-design (tight frame)
	fano7	Fano plane $S(2, 3, 7) = PG(2, 2)$	projective 2-design (tight frame)
	kuniform3	random 3-uniform blocks	arity sweep
	kuniform4	random 4-uniform blocks	arity sweep
	sunflower	shared core + petals	degenerate (shared hub)
extremal	petersen	$(3, 5)$ -cage, $SRG(10, 3, 0, 1) = K(5, 2)$	expander & tight frame
	kneser	$K(n, 2)$ disjointness graph	generalises Petersen
	grotzsch	Mycielskian $\mu(C_5)$	triangle-free, $\chi=4$, hierarchical
	expander	random d -regular	near-maximal spectral gap

Table 1: The zoo. Generic + extremal graphs build at the requested N (fixed- N graphs — Petersen, Grötzsch — ignore it and use their natural size); designs come at their star-expanded size.

2 Invariants (the walk-structure measures)

For each topology: algebraic connectivity (expansion floor), adjacency spectral gap (walk mixing), *frame coherence* (walk overlap; lower = more distinct walks), $\mathbb{Z}_2$ frustration (walk holonomy; -1 when $N > 18$ skips the brute-force), and degree irregularity.

topology	N	alg. conn.	spec. gap	coherence	frust.
chain	9	0.12	0.28	0.71	0
ring	9	0.47	0.47	0.50	0
star	9	1.00	2.83	1.00	0
tree	9	0.18	0.48	1.00	0
grid	9	1.00	1.41	0.71	2
small_world	9	0.55	0.92	0.82	2
random	9	0.61	1.61	0.67	1
complete	9	9.00	9.00	0.62	11
petersen	10	2.00	2.00	0.33	2
kneser(9,2)	36	20.0	20.0	0.52	—
grotzsch	11	2.23	2.70	0.67	3
expander	10	0.56	0.56	0.67	2
steiner9	21	1.70	1.73	0.33	—
fano7	14	1.59	1.59	0.33	0
kuniform3	17	0.39	0.97	0.71	0
kuniform4	17	0.69	1.94	0.75	0
sunflower	11	0.44	1.41	1.00	0

Table 2: Coherence is the discriminating axis: the $\mu=1$ designs/SRGs (Petersen, Steiner, Fano) reach the equiangular-tight-frame value $1/3$ (green); the hub/degenerate topologies (star, tree, sunflower) sit at 1.0 (red).

3 Use, and one caveat

The zoo feeds `controller_bench`: each topology becomes a learned HSiKAN controller fitting each plant. Invariants are the regressors. **One result already in hand (companion note for Kati):** frame coherence does *not* rank control — the matched (isomorphic) controller wins per plant, not the lowest-coherence one. So the zoo’s value is as the *family* over which the matching law is measured, not a ranking of “best” topologies. Code: `hymeko_rl/{topology_zoo, topology_invariants, hypergraph_...}` tests under `hymeko_rl/tests/`.

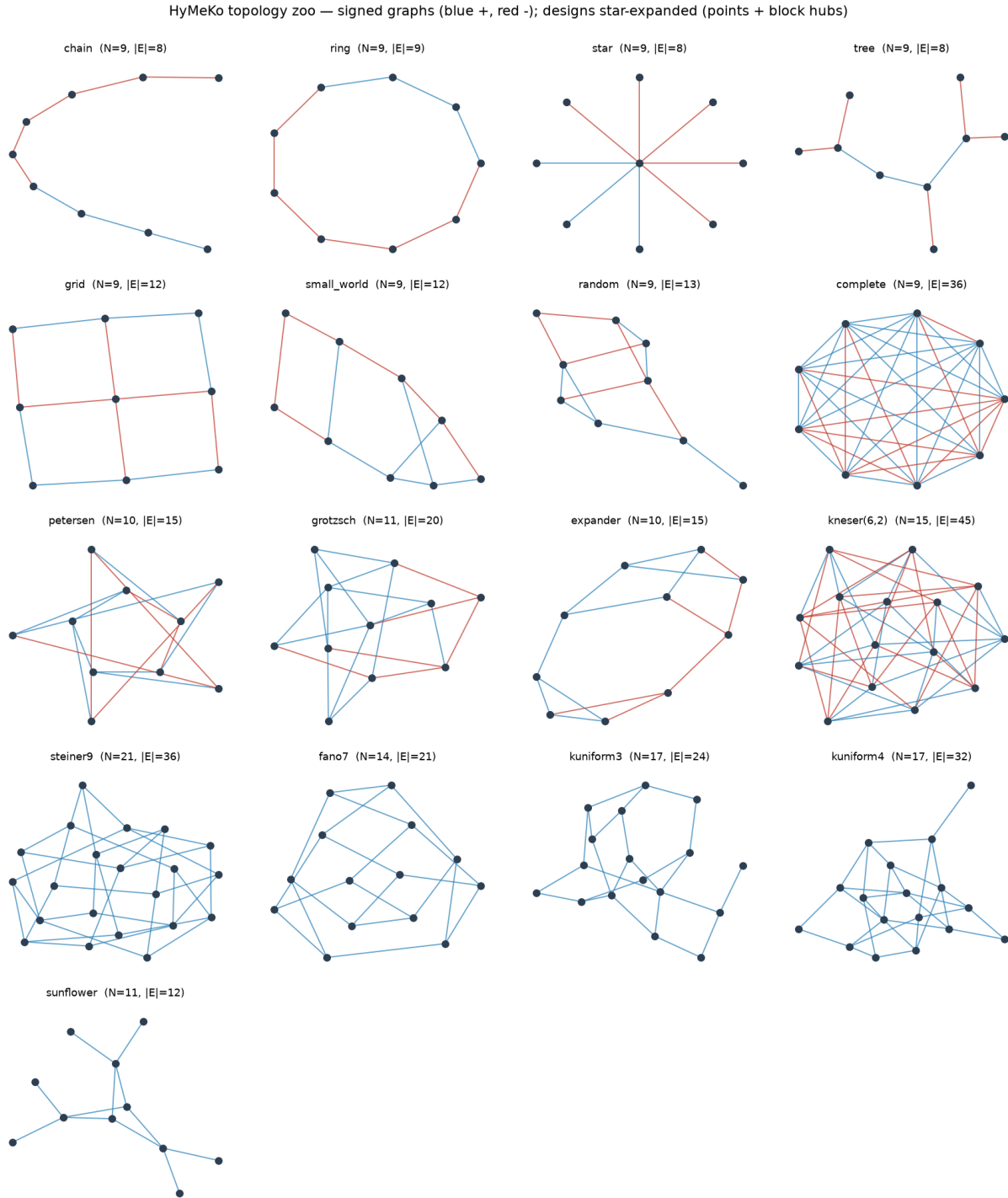


Figure 1: The zoo, drawn. Signed graphs (blue +, red -); designs are star-expanded (points + block hubs). Note Petersen's outer pentagon with inner pentagram, the central hubs of star / sunflower, and the dense block-structure of the Steiner / Fano designs.

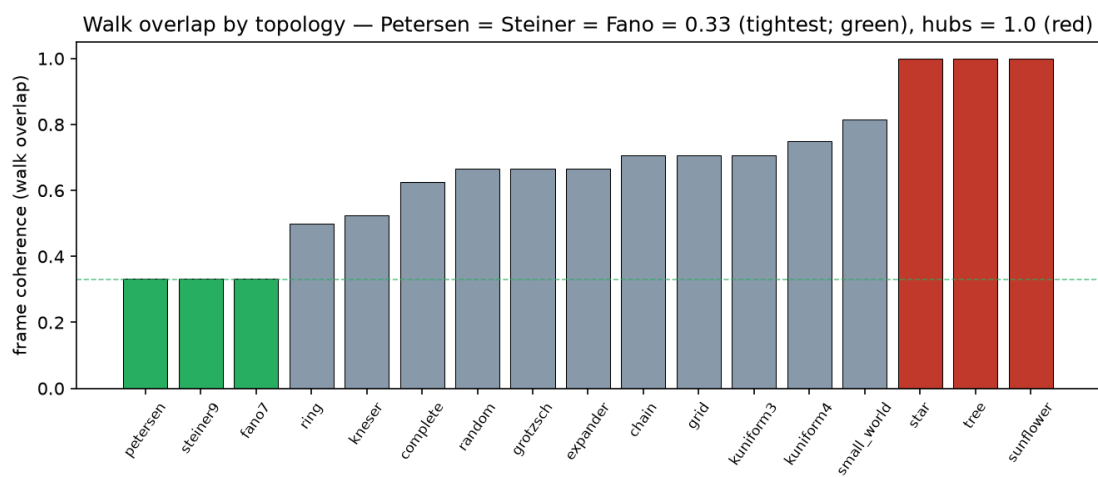


Figure 2: Walk overlap (frame coherence) by topology, sorted.